

PROJECT REPORT

Water Angel: Smart IoT-Based Water Management and Monitoring System

Submitted By

Name: Salman Khan

Roll No: 111

Program: BS Computer Science (Batch 12)

Semester: 4th (Section C)

Team Members

Member 1: Shayan Asim (Roll No: 93, Section B)

Member 2: Muhammad Hamza Jamal (Roll No: 57, Section B)

Submitted To

Supervisor: Dr. Saeed Ahmad

Designation: Assistant Professor

Department: Department of Computer Science

University: University of Swabi

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1. Introduction

Water is one of the most essential natural resources for human life, agriculture, and industrial activities. However, efficient water management remains a significant challenge, particularly in developing countries such as Pakistan. Rapid population growth, increasing water demand, and inefficient water usage practices have intensified concerns regarding water conservation and sustainable resource management. In many households, educational institutions, commercial buildings, and small industries, water storage tanks are still managed using traditional manual methods, which often result in water wastage, energy loss, and operational inefficiencies.

Traditional water tank systems typically require users to manually switch water pumps ON and OFF based on the tank's water level. This approach is highly dependent on human attention and can lead to common problems such as tank overflow, water shortages due to delayed pump activation, unnecessary electricity consumption, and potential damage to pumping equipment. Furthermore, conventional systems provide little or no information about water quality, pump performance, or water usage patterns, making effective monitoring and decision-making difficult.

With the advancement of the Internet of Things (IoT), cloud computing, and mobile technologies, it has become possible to develop intelligent systems capable of automating water management processes while providing real-time monitoring and remote accessibility. Smart water management solutions can significantly improve resource utilization, reduce operational costs, and enhance user convenience by integrating sensors, microcontrollers, cloud platforms, and mobile applications into a unified system.

To address these challenges, this project proposes **Water Angel**, a Smart IoT-Based Water Management and Monitoring System. Water Angel utilizes an ESP32 microcontroller, ultrasonic water level sensing, TDS-based water quality monitoring, cloud integration through Supabase, and a Progressive Web Application (PWA) to provide a comprehensive and intelligent water management solution. The system is designed using an offline-first architecture, allowing critical functions such as automatic pump control, water level monitoring, and water quality measurement to continue operating even in the absence of internet connectivity.

In addition to automated pump control, Water Angel provides users with the ability to remotely monitor and control the system through a mobile application, configure tank parameters and operational thresholds, update Wi-Fi credentials, receive alerts, and access pump usage analytics. By combining automation, IoT connectivity, cloud services, and data-driven insights, the proposed system aims to minimize water wastage, optimize energy consumption, improve water quality awareness, and enhance the overall efficiency of water management systems.

Therefore, Water Angel represents a practical, scalable, and intelligent solution that can be deployed in residential, educational, commercial, agricultural, and industrial environments to support modern water conservation and management practices.

2. Problem Statement

Traditional water tank management systems face several critical limitations that result in inefficient water usage, energy loss, and lack of monitoring:

- Water pumps are operated manually, leading to human errors such as forgetting to turn the pump ON or OFF
- Tank overflow occurs due to delayed switching, causing wastage of clean water
- Underflow conditions happen when the pump is not turned ON in time, resulting in water shortage
- Continuous or improper pump usage leads to unnecessary electricity consumption
- Lack of automation increases the risk of pump damage due to dry running conditions
- No real-time monitoring of water level is available in traditional systems
- Users cannot remotely check or control pump status when they are away
- Absence of data analytics prevents understanding of water usage patterns and pump behavior
- No intelligent decision-making or smart control system exists in conventional setups
- Water quality is not monitored, and users are unaware of contamination or high TDS levels
- No alert or notification system exists for abnormal conditions such as overflow, dry run, or poor water quality
- Overall, traditional systems lack integration of IoT, automation, and smart monitoring features

3. Proposed Solution Overview

To overcome the limitations of traditional systems, this project proposes Water Angel, an IoT-based smart water management system. The system uses an ESP32 microcontroller to automate pump control based on real-time water level detection and predefined thresholds.

The system operates in an offline-first mode, ensuring continuous functionality even without internet connectivity. Additionally, it integrates a mobile application (PWA) that allows users to remotely monitor and control the system, configure settings such as tank height and thresholds, and update Wi-Fi credentials.

A cloud backend using Supabase is used to synchronize data, store configurations, and enable analytics and alert generation. The system also includes water quality monitoring using a TDS sensor and provides pump usage analytics for better decision-making.

The developed system has been successfully implemented and tested. It provides automatic pump control, water level monitoring, TDS-based water quality assessment, cloud synchronization, and remote monitoring through a Progressive Web Application (PWA). The system continues to perform critical functions even when internet connectivity is unavailable, ensuring reliable operation under real-world conditions.

Technology Stack

- ESP32 – Main microcontroller handling sensor reading, pump control, and Wi-Fi connectivity
- Supabase – Cloud backend (PostgreSQL) for data storage, real-time sync, and authentication
- PWA App – Mobile application for remote monitoring, control, and configuration

4. Objectives of the Project

- To design an automated water pump control system using real-time water level sensing
- To prevent water wastage by avoiding overflow and underflow conditions
- To monitor and display real-time water level information in percentage form. To measure water quality using a TDS sensor
- To enable remote monitoring and control through a mobile application (PWA)
- To allow users to set tank height and threshold values from the mobile application
- To provide manual pump control while maintaining automatic mode operation
- To enable Wi-Fi credential configuration through the mobile application
- To implement offline-first system functionality for uninterrupted operation
- To integrate cloud-based backend services using Supabase for data synchronization
- To generate real-time analytics and intelligent alerts for enhanced monitoring and decision-making.

5. System Architecture

The Water Angel system consists of two major components: the Hardware Architecture and the Software Architecture. The hardware layer is responsible for sensing, monitoring, and controlling water-related operations, while the software layer provides remote monitoring,

configuration, analytics, and cloud synchronization. Together, these components create a reliable and intelligent water management system capable of operating both online and offline.

5.1 Hardware Architecture

The hardware architecture of Water Angel is centered around the ESP32 microcontroller, which acts as the main processing and control unit. Various sensors and actuators are connected to the ESP32 to enable real-time monitoring and automatic pump control.

- **ESP32 Microcontroller**

The ESP32 serves as the brain of the system. It reads sensor data, performs water level calculations, executes pump control logic, manages Wi-Fi communication, stores configuration settings in EEPROM, and synchronizes data with the cloud backend.

- **Ultrasonic Sensor**

The ultrasonic sensor is installed at the top of the water tank to measure the distance between the sensor and the water surface. Based on the configured tank height, the system calculates the water level percentage and uses this information for automatic pump control and monitoring.

- **TDS Sensor**

The Total Dissolved Solids (TDS) sensor is used to monitor water quality. It measures the concentration of dissolved substances in water and helps identify poor-quality water conditions. When the measured TDS value exceeds predefined limits, the system can generate alerts for the user.

- **Relay Module**

The relay module acts as an electronic switch that allows the ESP32 to safely control the water pump. Based on the current water level and configured thresholds, the relay automatically turns the pump ON or OFF.

- **Water Pump**

The water pump is responsible for filling the water tank. It can operate in both automatic and manual modes. In automatic mode, the pump is controlled by the system based on water level thresholds, while in manual mode it can be controlled remotely through the PWA application.

- **LCD Display**

A 16×2 I2C LCD is used to provide local real-time information. It displays water level percentage, TDS values, pump status, system mode, and other important operating information.

- **LEDs and Buzzer**

LED indicators and a buzzer provide visual and audible alerts. The green LED indicates normal operation and pump activity, while the red LED and buzzer are used to indicate critical conditions such as low water level or poor water quality.

- **EEPROM Storage**

EEPROM memory is used to store important system configurations, including Wi-Fi credentials, tank height, calibration data, and threshold settings. This ensures that the system retains its configuration even after power loss or restart.

- **Setup Button and Wi-Fi Provisioning**

A dedicated push button is integrated into the hardware to simplify Wi-Fi configuration. When the button is pressed and held for approximately three seconds, the ESP32 enters Setup Mode and automatically creates a Wi-Fi hotspot named "Water Angel".

Users can connect their mobile device to this hotspot and access the Water Angel PWA configuration page. Through this interface, users can securely enter their Wi-Fi network credentials. Once submitted, the credentials are stored in EEPROM memory, and the ESP32 automatically attempts to connect to the specified Wi-Fi network.

This approach eliminates the need for hardcoded Wi-Fi credentials and allows users to easily reconfigure the device whenever network settings change.

- **Offline-First Hardware Operation**

- The Water Angel system is designed using an offline-first architecture. Critical functionalities such as water level monitoring, TDS measurement, threshold processing, pump control, LCD updates, LED indicators, and alarm generation are executed locally by the ESP32 without requiring an internet connection.
- Configuration settings, including Wi-Fi credentials, tank height, and threshold values, are stored in EEPROM memory to ensure that the system continues operating normally even after power failures or internet outages.
- This design guarantees uninterrupted operation and reliability in environments with unstable network connectivity.

5.2 Software Architecture

The software architecture of Water Angel consists of a Progressive Web Application (PWA), a cloud backend powered by Supabase, and an offline-first firmware running on the ESP32. These components work together to provide remote accessibility, data storage, analytics, and intelligent monitoring.

- **Water Angel PWA Application**

The Water Angel Progressive Web Application (PWA) serves as the primary user interface. It allows users to monitor water levels, view water quality information, control the pump remotely, configure thresholds, set tank height, update Wi-Fi credentials, view analytics, and receive system alerts.

- **Supabase Backend**

Supabase is used as the cloud backend for the system. It provides database services, real-time synchronization, authentication, and secure data storage. Sensor readings, pump status, user configurations, and analytics data are stored and managed through Supabase.

- **Offline-First Operation**

One of the key features of Water Angel is its offline-first architecture. Critical functionalities such as water level monitoring, automatic pump control, threshold processing, and water quality monitoring continue to operate even when internet connectivity is unavailable. This ensures reliable operation under all conditions.

- **Cloud Synchronization**

When internet connectivity is available, the ESP32 synchronizes sensor readings, pump status, and configuration data with the Supabase cloud database. This allows users to remotely monitor and manage the system through the PWA application.

- **Analytics and Alert System**

The analytics module collects operational data such as water level trends, pump usage statistics, and water quality information. This data is processed and presented through the PWA application to help users understand usage patterns and make informed decisions. The system also generates alerts for critical situations such as low water level, high TDS values, and abnormal pump behavior.

- **Automatic Mode**

In Automatic Mode, the ESP32 controls the water pump autonomously based on user-defined threshold values. The system continuously monitors the water level and compares it against the configured upper and lower thresholds.

When the water level falls below the lower threshold, the pump is automatically switched ON. When the water level reaches the upper threshold, the pump is automatically switched OFF. This prevents both water overflow and underflow while minimizing user intervention.

Automatic Mode ensures efficient water management and optimal energy utilization.

- **Manual Mode**

In Manual Mode, users can directly control the pump through the Water Angel PWA application. Commands sent from the application are synchronized through the Supabase backend and executed by the ESP32.

This mode allows users to switch the pump ON or OFF remotely whenever required while still benefiting from real-time monitoring and system alerts.

Manual Mode is particularly useful for maintenance activities, emergency situations, or special operational requirements.

6. Hardware Implementation

The hardware implementation of Water Angel consists of an ESP32-based control unit integrated with multiple sensors, indicators, memory components, and a relay-controlled water pump. The hardware is designed to provide reliable water level monitoring, water quality assessment, automatic pump control, local user feedback, and wireless connectivity for cloud synchronization and remote management.

6.1 Circuit Description

The Water Angel system is built around the ESP32 microcontroller, which serves as the central processing and communication unit. All sensors, indicators, and control devices are connected directly to the ESP32.

The ultrasonic sensor is mounted at the top of the water tank and is connected to the ESP32 through dedicated Trigger and Echo pins. It continuously measures the distance between the sensor and the water surface, allowing the system to calculate the current water level percentage.

The TDS sensor is connected to an analog input pin of the ESP32 and is used to measure the quality of water by estimating the concentration of dissolved solids.

A relay module is connected to one of the digital output pins of the ESP32. The relay acts as an electrically isolated switch that controls the water pump based on commands generated by the control logic.

A 16×2 I2C LCD is connected through the I2C communication interface and is used to display water level information, TDS values, pump status, and system operating mode.

Green and red LEDs are used as visual indicators. The green LED indicates normal system operation and pump activity, while the red LED indicates warning or critical conditions. A buzzer is also connected to provide audible alerts during abnormal situations.

A push button is connected to the ESP32 and is used to activate Setup Mode. When pressed and held for approximately three seconds, the ESP32 creates a Wi-Fi hotspot named "**Water Angel**", allowing users to configure Wi-Fi credentials through the PWA setup interface.

Configuration settings such as Wi-Fi credentials, tank height, and threshold values are stored in the ESP32's EEPROM memory to ensure persistence after power interruptions.

6.2 Working Principle

The operation of Water Angel begins when the ESP32 powers up and loads all previously saved configuration parameters from EEPROM memory. These parameters include tank height, water level thresholds, calibration information, and Wi-Fi credentials.

After initialization, the ultrasonic sensor continuously measures the distance to the water surface. The ESP32 processes this distance measurement and converts it into a water level percentage based on the configured tank height.

Simultaneously, the TDS sensor measures water quality and provides real-time TDS values to the system.

The calculated water level is compared against predefined threshold values. In Automatic Mode, if the water level falls below the lower threshold, the ESP32 activates the relay and turns the water pump ON. When the water level reaches the upper threshold, the pump is automatically switched OFF.

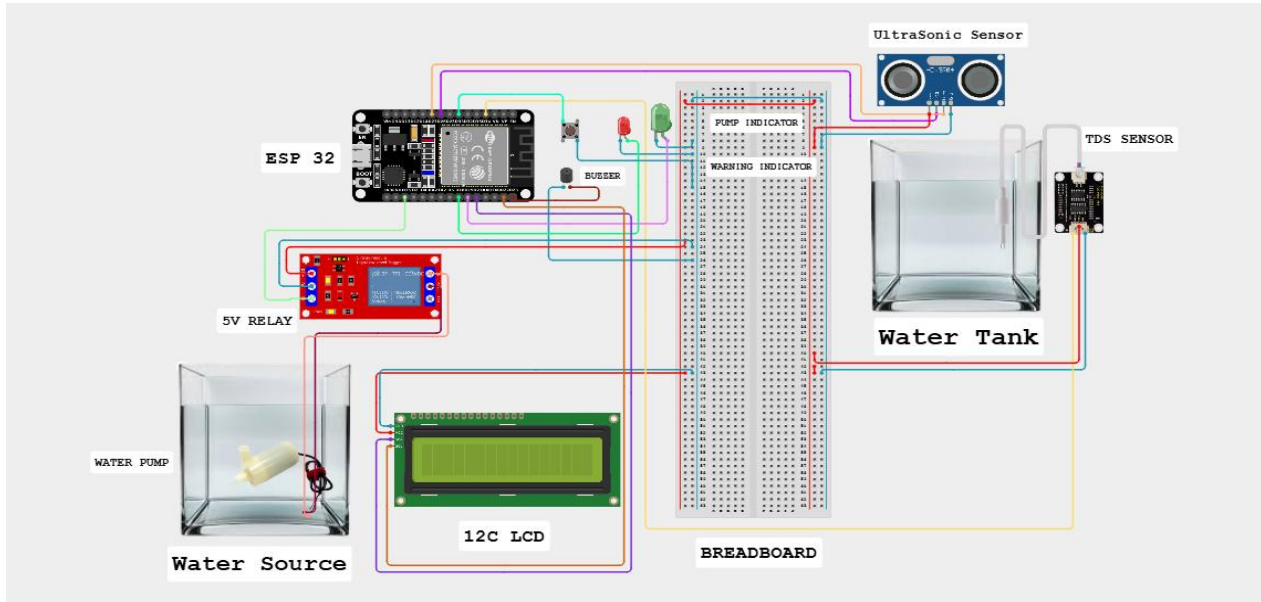
In Manual Mode, pump control commands are received through the Water Angel PWA application and executed by the ESP32 while monitoring functions continue normally.

The LCD continuously displays system information, while LEDs and the buzzer provide visual and audible feedback regarding pump operation and critical conditions.

When internet connectivity is available, the ESP32 synchronizes sensor readings, pump status, analytics data, and configuration settings with the Supabase cloud backend. Users can then monitor and control the system remotely through the Water Angel PWA.

If internet connectivity is lost, the system continues to operate normally using its offline-first architecture. Automatic pump control, water level monitoring, water quality monitoring, and alert generation remain fully functional until connectivity is restored.

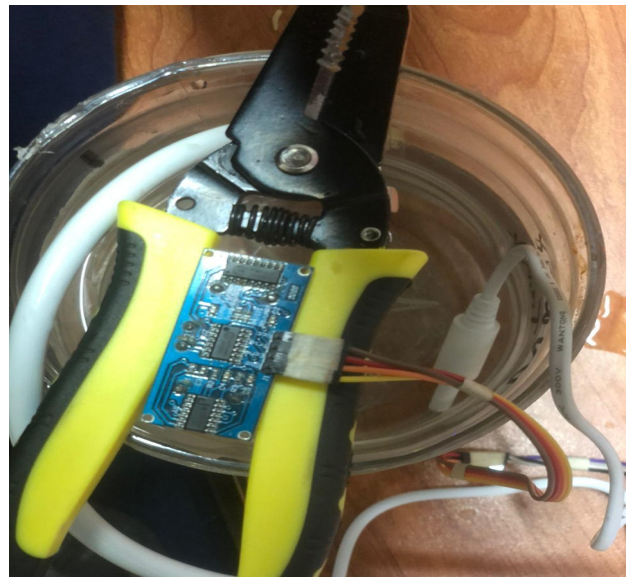
6.3 Hardware Images



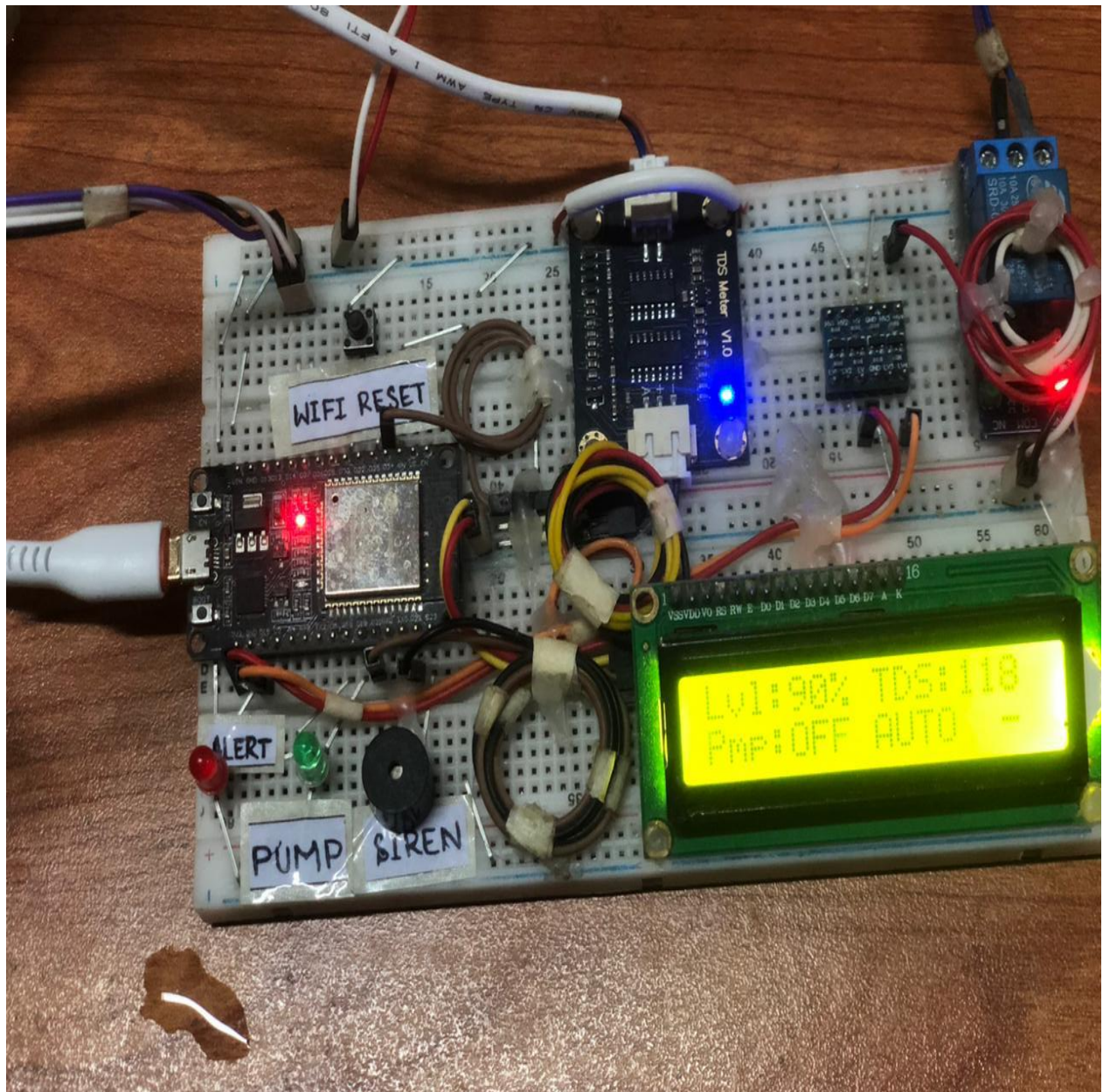
Circuit Diagram



Water Source



Water Tank



Main Circuit Board

7. Water Angel PWA Application

7.1 App Overview

The Water Angel Progressive Web Application (PWA) serves as the primary user interface for monitoring and controlling the smart water management system. The application enables users to remotely access real-time water tank information, control pump operations, configure system parameters, view analytics, and receive alerts from any device with a web browser.

The PWA communicates with the Supabase cloud backend and the ESP32 controller to provide a seamless user experience. It is designed with a simple and intuitive interface that allows users to efficiently manage water resources while monitoring system performance in real time.

The application consists of five main sections: Home, Analytics, Control, Alerts, and Profile. Each section is designed to provide specific functionality for system monitoring, configuration, and management.

7.2 Key Features

- Real-time water level monitoring
- Water quality monitoring using TDS values
- Remote pump control
- Automatic and manual operation modes
- Tank height configuration
- Water level threshold configuration
- Wi-Fi credential management
- Historical data visualization
- Pump analytics and usage insights
- Alert and notification system
- Cloud synchronization through Supabase
- Offline-first hardware operation

7.3 Application Screenshots and Functionality

• Tab 1: Home Dashboard

Description

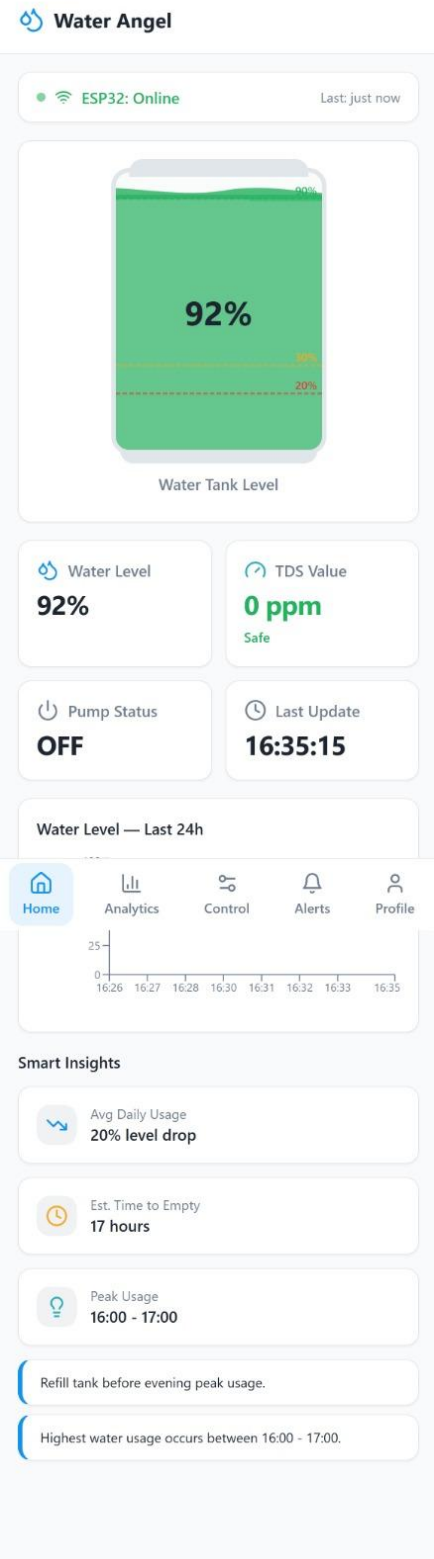
The Home Dashboard serves as the central monitoring screen of the Water Angel application. It provides users with a real-time overview of the complete water management system and displays the most important operational information in a single interface

Features Available on the Home Dashboard

- Device Connectivity Status
- Visual Water Tank Representation
- Water Level Monitoring
- Water Quality Monitoring
- Pump Status Monitoring
- Water Level History Graph
- Smart Insights

App URL:

<https://preview--water-angel.lovable.app/>



• Tab 2: Analytics

Description

The Analytics Dashboard provides users with historical data visualization, performance analysis, water consumption insights, and intelligent recommendations. It transforms raw sensor readings into meaningful information that helps users understand water usage patterns, monitor system performance, and make informed decisions regarding water management.

The dashboard continuously analyzes water level data, TDS measurements, and pump activity collected by the ESP32 and stored in the Supabase backend.

Features Available on the Home Dashboard

- Water Level History
- Daily Water Consumption Analysis(7 Days)
- Water Quality (TDS) Trend Analysis(7 Days)
- Pump Activity Monitoring
- Data Insights Panel
- Water Efficiency Score
- Weekly Smart Report
- Recommendations



• Tab 3: Controls

Description

The Control Dashboard provides users with complete control over the operation and configuration of the Water Angel system. Through this interface, users can select the pump operating mode, manually control pump operation, configure tank dimensions, and define water level thresholds used by the automatic control algorithm.

The dashboard acts as the primary configuration center of the system and enables customization according to different tank sizes and user requirements.

Pump Operating Modes

Water Angel supports two operational modes:

1 Automatic Mode

- Fully automated operation
- Prevents overflow and underflow conditions
- Saves electricity
- Minimizes user involvement
- Ensures reliable operation even during internet outages

2. Manual Mode

- Immediate pump control
- Useful during maintenance activities
- Allows emergency operation
- Provides flexibility for special situations

Other Features

- Tank Height Configuration
- Threshold Configuration

Pump Control

Automatic

Pump Control

Manual

• Tab 4: Alerts

Description

The Alerts Dashboard is responsible for notifying users about important system events, abnormal operating conditions, and potential risks detected by the Water Angel system. The dashboard continuously analyzes water level measurements, water quality readings, and system behavior to generate real-time alerts whenever predefined conditions are triggered.

Each alert contains a detailed description of the detected issue, its category, and the exact date and time of occurrence. This allows users to quickly identify problems and take corrective action before they result in water wastage, equipment damage, or water quality concerns.

Features Available on the Home Dashboard

- Critical Water Level Alerts
- High TDS Alerts
- Leak Detection Alerts
- Alert Categorization

Benefits of the Alerts

The Alerts Dashboard significantly improves system reliability and user awareness by providing early warnings about critical situations. Instead of relying solely on manual monitoring, users receive immediate notifications regarding water shortages, water quality concerns, and potential leakages.

The alert system contributes to:

- Reduced water wastage
- Early leak detection
- Improved water quality awareness
- Enhanced system reliability
- Better maintenance planning

• Tab 5: profile

Alerts

Sudden water level drop detected: 83.5% decrease. Possible leak.
Water_level • 17/06/2026, 16:51:12

Critical water level: 9.6%. Immediate refill required.
Water_level • 17/06/2026, 16:51:12

Sudden water level drop detected: 85.3% decrease. Possible leak.
Water_level • 17/06/2026, 16:38:47

Critical water level: 2.8%. Immediate refill required.
Water_level • 17/06/2026, 16:38:47

Sudden water level drop detected: 52.0% decrease. Possible leak.
Water_level • 17/06/2026, 16:37:47

High TDS: 1099 ppm. Water quality unsafe.
Water_quality • 17/06/2026, 16:30:12

Sudden water level drop detected: 15.3% decrease. Possible leak.
Water_level • 17/06/2026, 16:28:53

Description

The Profile Dashboard serves as the account management and system configuration center of the Water Angel application. It allows users to manage their account information, pair hardware devices, configure notification preferences, update system settings, and perform account-related operations.

This dashboard ensures that users can easily personalize their experience and manage both the application and connected hardware devices from a single interface.

Features Available on the Profile Dashboard

- User Information
- Device Pairing System

Each Water Angel hardware unit is assigned a unique Device Key during registration.

The Device Pairing section allows users to connect the PWA application with a specific Water Angel device by entering its unique Device Key.

This mechanism ensures secure communication between the user account and the designated hardware device.

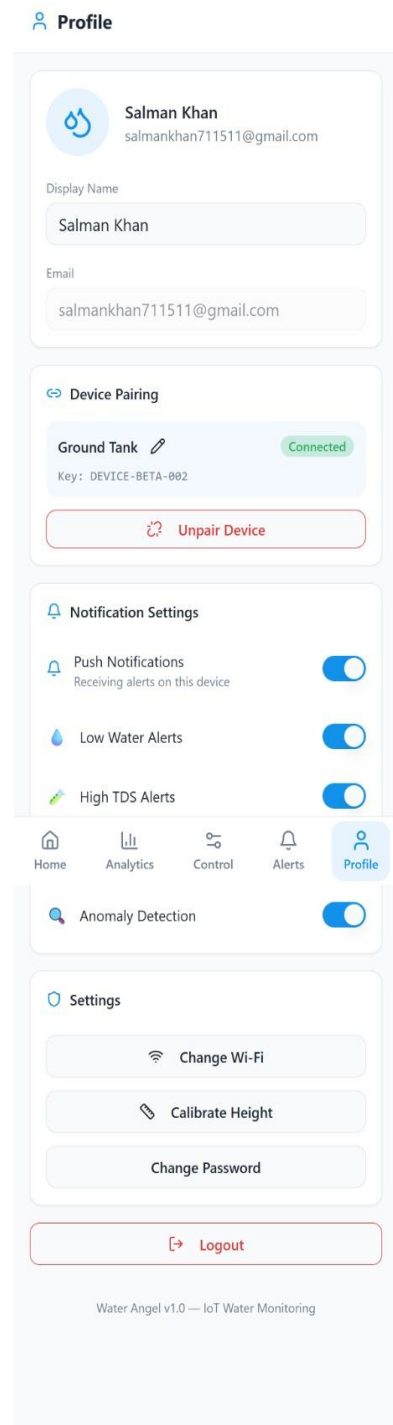
Notification Preferences

- Low Water Level Alerts
- High TDS Alerts
- Pump Status Alerts
- Anomaly Detection Alerts

System Settings

- Change Wi-Fi Configuration
- Tank Height Calibration
- Change Password

Logout Functionality



8. Project Workflow

The Water Angel system follows a structured workflow that enables intelligent water management through continuous monitoring, automated decision-making, cloud synchronization, and remote user interaction. The workflow begins when the system is powered ON and the ESP32 initializes all stored configurations, including Wi-Fi credentials, tank height, and threshold values from EEPROM memory.

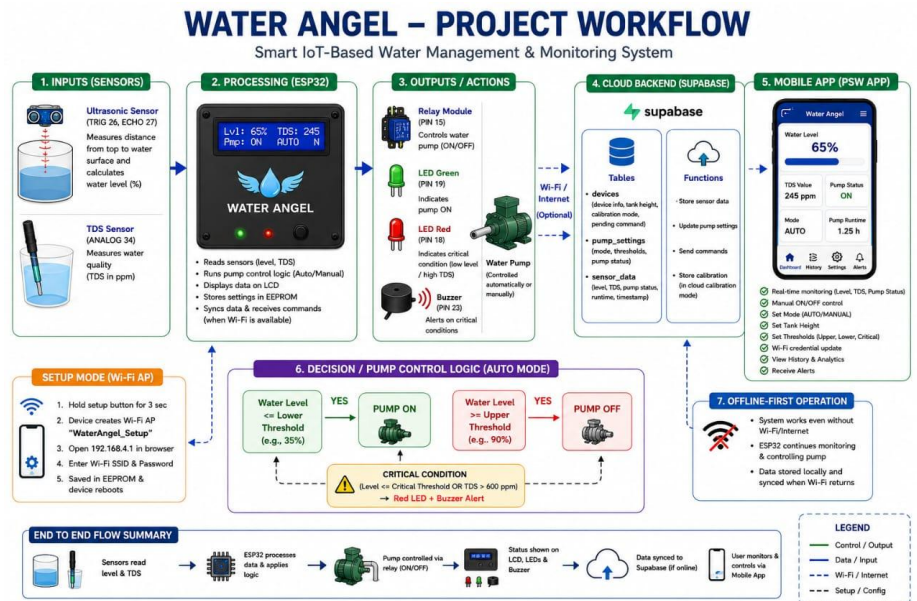
The system then continuously monitors water level and water quality using the ultrasonic sensor and TDS sensor. Based on the measured values and user-defined thresholds, the ESP32 determines whether the water pump should be activated or deactivated.

In Automatic Mode, the pump is controlled autonomously according to the configured upper and lower water level thresholds. In Manual Mode, users can remotely control the pump through the Water Angel PWA application.

The system updates local indicators such as the LCD display, LEDs, and buzzer to provide real-time operational feedback. When internet connectivity is available, sensor readings, pump status analytics information, and configuration settings are synchronized with the Supabase cloud backend. Users can then access this information through the Water Angel PWA for remote monitoring and management.

If internet connectivity becomes unavailable, the system continues to perform all critical operations locally using its offline-first architecture. Once connectivity is restored, synchronization with the cloud resumes automatically.

Additionally, if the user presses and holds the setup button for approximately three seconds, the ESP32 enters Setup Mode and creates a Wi-Fi hotspot named "Water Angel". Users can connect to this hotspot and configure Wi-Fi credentials through the PWA interface. The credentials are stored in EEPROM and used for future network connections.



9. Benefits of the Project

- Reduces water wastage by preventing overflow conditions
- Saves electricity through optimized pump operation
- Prevents pump damage caused by dry running
- Provides real-time monitoring of water level and quality
- Enables remote control and monitoring through mobile application
- Improves user convenience by automating manual tasks
- Supports smart decision-making through analytics and usage data
- Provides early alerts for abnormal conditions (overflow, dry run, poor water quality)
- Enhances safety by monitoring water quality (TDS levels)
- Works both online and offline for maximum reliability

10. Project Deployment Areas

- Residential houses and domestic water tanks
- Apartment buildings and housing societies
- Educational institutions (universities, colleges, hostels)
- Commercial buildings (offices, malls, complexes)
- Agricultural farms and irrigation systems
- Small industries requiring water management systems
- Public water supply and community storage systems

11. Results and Achievements

The Water Angel project was successfully designed, developed, and tested as a complete IoT-based smart water management system. The project achieved its primary objectives of automating water pump operation, monitoring water quality, providing remote accessibility, and enabling intelligent decision-making through analytics and alerts.

The following key achievements were accomplished during the development and testing phases:

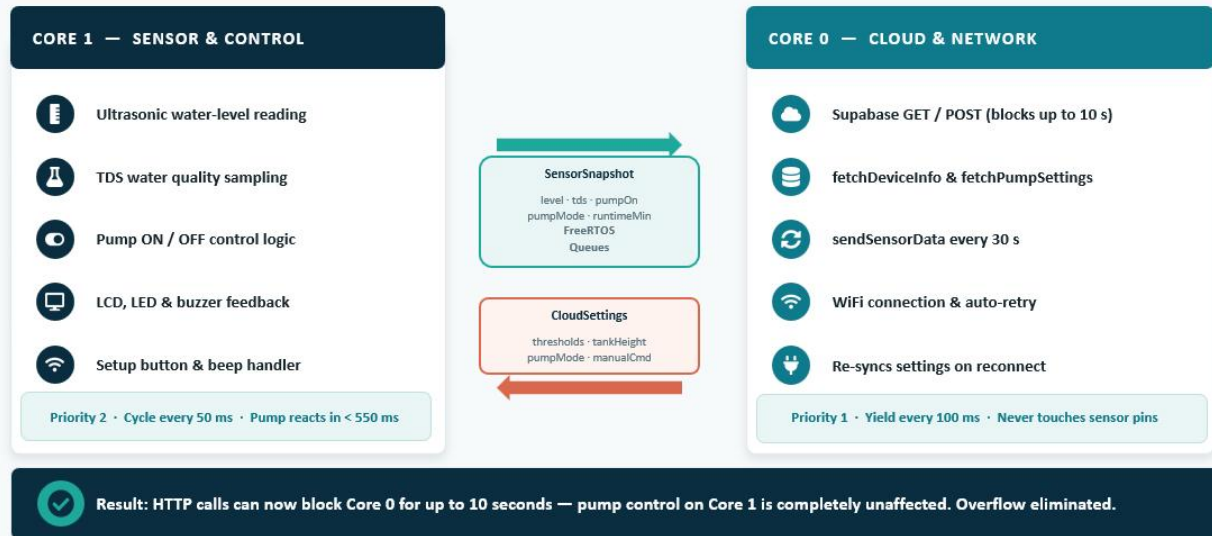
- Successful Automatic Pump Control
- Offline-First Operation Verified
- Cloud Synchronization Successfully Tested
- Real-Time Monitoring Implemented

- Water Quality Monitoring Implemented
- Alert and Notification System Successfully Implemented
- Intelligent Analytics Dashboard Developed
- Wi-Fi Provisioning and Remote Configuration Implemented
- Progressive Web Application Successfully Developed

12. Dual-Core Architecture

Dual-Core FreeRTOS Architecture

HTTP calls to Supabase were blocking the sensor loop — causing overflow. Splitting across two cores eliminated the problem entirely.



13. Future Enhancements

- AI-Based Water Consumption Prediction
- Solar-Powered Operation
- Mobile Push Notifications
- Advanced Water Quality Monitoring
- Smart Community Water Management

- Leakage Detection

14. Conclusion

The Water Angel project aims to modernize traditional water management systems by integrating IoT, automation, and cloud-based technologies. It provides an efficient, reliable, and intelligent solution for water monitoring and pump control. The system not only reduces water and electricity wastage but also introduces smart analytics, remote accessibility, and water quality monitoring, making it suitable for both domestic and industrial applications.